Supplementary Material for "Similarity and Correlation Learning for Multitask Optimization"

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TABLE S.I PROPERTIES OF CEC17 MULTITASK TEST SUITE

Problem	Task	Dimension	Degree of Intersection	Inter-task Similarity
CIHS	Griewank (T_1)	50	Complete intersection	1.0000
	Rastrigin (T_2)	50		
CIMS	Ackley (T_1)	50	Complete intersection	0.2261
	Rastrigin (T_2)	50		
CILS	Ackley (T_1)	50	Complete intersection	0.0002
	Schwefel (T_2)	50		
PIHS	Rastrigin (T_1)	50	Partial intersection	0.8670
	Sphere (T_2)	50		
PIMS	Ackley (T_1)	50	Partial intersection	0.2154
	Rosenbrock (T_2)	50		
PILS	Ackley (T_1)	50	Partial intersection	0.0725
	Weierstrass (T_2)	25		
NIHS	Rosenbrock (T_1)	50	No intersection	0.9434
	Rastrigin (T_2)	50		
NIMS	Griewank (T_1)	50	No intersection	0.3669
	Weierstrass (T_2)	50		
NILS	Rastrigin (T_1)	50	No intersection	0.0016
	Schwefel (T_2)	50		

TABLE S.II RESULTS OF SCLMTO AND COMPARED ALGORITHMS ON PKAC PROBLEMS

Problem	SCLMTO	MFEA	MFEA-AKT	ASCMFDE	MTES	MKTDE	BoKT-DE	BoKT-GA
D5-T1	**3.50E−1**	3.50E−1(+)	3.50E−1(+)	3.50E−1(+)	3.51E−1(+)	3.50E−1(+)	3.50E−1(+)	3.50E−1(+)
D5-T2	2.16E−1	**2.16E−1(−)**	2.16E−1(−)	2.16E−1(+)	2.18E−1(+)	2.16E−1(−)	2.16E−1(−)	**2.16E−1(−)**
D10-T1	2.67E−1	2.67E−1(+)	2.67E−1(+)	2.67E−1(+)	2.69E−1(+)	2.67E−1(≈)	2.67E−1(+)	2.67E−1(+)
D10-T2	**2.93E−1**	2.93E−1(+)	2.93E−1(+)	2.94E−1(+)	3.18E−1(+)	2.93E−1(+)	2.93E−1(+)	2.93E−1(+)
D15-T1	**2.95E−1**	2.95E−1(+)	2.95E−1(+)	3.00E−1(+)	3.49E−1(+)	2.95E−1(+)	2.95E−1(+)	2.95E−1(+)
D15-T2	2.07E−1	2.07E−1(+)	2.07E−1(+)	2.07E−1(+)	2.13E−1(+)	2.07E−1(≈)	2.07E−1(≈)	2.07E−1(+)
D20-T1	3.57E−1	3.57E−1(+)	3.57E−1(+)	3.57E−1(+)	3.63E−1(+)	3.57E−1(+)	**3.57E−1(−)**	3.57E−1(≈)
D20-T2	2.46E−1	2.46E−1(+)	2.46E−1(+)	2.60E−1(+)	3.38E−1(+)	2.46E−1(+)	2.46E−1(+)	2.46E−1(+)
D30-T1	3.00E−1	3.00E−1(+)	3.00E−1(+)	3.21E−1(+)	3.94E−1(+)	3.00E−1(+)	3.01E−1(+)	3.00E−1(+)
D30-T2	2.10E−1	2.10E−1(+)	2.10E−1(+)	2.11E−1(+)	2.46E−1(+)	2.10E−1(+)	2.10E−1(−)	**2.10E−1(−)**
D50-T1	2.73E−1	2.73E−1(≈)	2.73E−1(+)	2.75E−1(+)	3.24E−1(+)	2.73E−1(+)	2.73E−1(+)	**2.73E−1(−)**
D50-T2	3.07E−1	3.09E−1(+)	3.09E−1(+)	3.44E−1(+)	4.19E−1(+)	3.09E−1(+)	3.10E−1(+)	3.08E−1(+)
D80-T1	1.85E−1	1.85E−1(+)	1.85E−1(+)	1.92E−1(+)	3.07E−1(+)	1.85E−1(+)	1.85E−1(+)	**1.85E−1(≈)**
D80-T2	4.13E−1	4.15E−1(+)	4.15E−1(+)	4.39E−1(+)	4.80E−1(+)	4.15E−1(+)	4.16E−1(+)	4.14E−1(+)
D100-T1	**1.86E−1**	1.86E−1(+)	1.86E−1(+)	1.95E−1(+)	3.31E−1(+)	1.86E−1(+)	1.86E−1(+)	1.86E−1(≈)
D100-T2	4.15E−1	4.17E−1(+)	4.17E−1(+)	4.43E−1(+)	4.85E−1(+)	4.17E−1(+)	4.18E−1(+)	4.16E−1(+)
Number of + / ≈ / −		14 / 1 / 1	15 / 0 / 1	16 / 0 / 0	16 / 0 / 0	13 / 2 / 1	12 / 1 / 3	10 / 3 / 3
Problem	**SCLMTO**	**RLMFEA**	**BLKT-DE**	**EMTO-AI**	**MFEA-DGD**	**SSLT-DE**	**SSLT-GA**	**MMLMTO**
D5-T1	**3.50E−1**	3.50E−1(+)	3.50E−1(−)	3.50E−1(+)	3.50E−1(+)	3.50E−1(+)	3.50E−1(+)	3.50E−1(+)
D5-T2	2.16E−1	2.16E−1(−)	2.16E−1(−)	2.16E−1(+)	2.16E−1(−)	2.16E−1(−)	2.16E−1(−)	2.16E−1(≈)
D10-T1	2.67E−1	2.67E−1(+)	**2.67E−1(−)**	2.67E−1(+)	2.68E−1(+)	2.67E−1(+)	2.67E−1(+)	2.67E−1(+)
D10-T2	**2.93E−1**	2.93E−1(+)	2.93E−1(+)	2.93E−1(+)	2.93E−1(+)	2.98E−1(+)	2.94E−1(+)	2.93E−1(+)
D15-T1	**2.95E−1**	2.95E−1(+)	2.95E−1(+)	2.98E−1(+)	**2.95E−1(≈)**	3.09E−1(+)	2.99E−1(+)	2.95E−1(+)
D15-T2	2.07E−1	2.07E−1(+)	**2.07E−1(−)**	2.07E−1(+)	2.08E−1(+)	2.07E−1(+)	2.07E−1(+)	2.07E−1(+)
D20-T1	3.57E−1	3.57E−1(+)	3.57E−1(≈)	3.57E−1(+)	3.58E−1(+)	3.57E−1(+)	3.57E−1(+)	3.57E−1(+)
D20-T2	2.46E−1	2.47E−1(+)	2.46E−1(+)	2.56E−1(+)	**2.46E−1(−)**	2.74E−1(+)	2.58E−1(+)	2.47E−1(+)
D30-T1	3.00E−1	3.02E−1(+)	3.00E−1(+)	3.17E−1(+)	**3.00E−1(−)**	3.35E−1(+)	3.15E−1(+)	3.02E−1(+)
D30-T2	2.10E−1	2.10E−1(+)	2.10E−1(+)	2.10E−1(+)	2.14E−1(+)	2.11E−1(+)	2.10E−1(+)	2.10E−1(+)
D50-T1	2.73E−1	2.73E−1(+)	2.73E−1(+)	2.74E−1(+)	2.78E−1(+)	2.76E−1(+)	2.74E−1(+)	2.73E−1(+)
D50-T2	3.07E−1	3.12E−1(+)	3.07E−1(≈)	3.47E−1(+)	**3.06E−1(−)**	3.60E−1(+)	3.33E−1(+)	3.13E−1(+)
D80-T1	1.85E−1	1.86E−1(+)	1.85E−1(+)	1.93E−1(+)	1.97E−1(+)	1.97E−1(+)	1.88E−1(+)	1.87E−1(+)
D80-T2	4.13E−1	4.18E−1(+)	4.13E−1(−)	4.43E−1(+)	**4.11E−1(−)**	4.48E−1(+)	4.31E−1(+)	4.21E−1(+)
D100-T1	**1.86E−1**	1.86E−1(+)	1.86E−1(+)	1.99E−1(+)	1.97E−1(+)	2.08E−1(+)	1.90E−1(+)	1.89E−1(+)
D100-T2	4.15E−1	4.19E−1(+)	4.14E−1(≈)	4.49E−1(+)	**4.11E−1(−)**	4.53E−1(+)	4.35E−1(+)	4.26E−1(+)
Number of + / ≈ / −		15 / 0 / 1	8 / 3 / 5	16 / 0 / 0	9 / 1 / 6	15 / 0 / 1	15 / 0 / 1	15 / 1 / 0

TABLE S.III EXPERIMENTAL RESULTS OF SCLMTO AND ITS VARIANTS

Problem	SCLMTO	SCLMTO-ktp(0.0)	SCLMTO-ktp(0.2)	SCLMTO-ktp(0.3)	SCLMTO-ktp(0.4)	SCLMTO-ktp(0.5)
CIHS-T1	**0**	2.55E−3(+)	**0(≈)**	5.87E−4(≈)	2.10E−3(+)	1.84E−3(+)
CIHS-T2	**0**	3.06E+2(+)	1.14E−14(+)	1.59E+0(+)	9.92E+0(+)	5.31E+0(+)
CIMS-T1	**1.49E−10**	2.59E−1(+)	1.53E−9(+)	1.57E−1(+)	4.37E−1(+)	3.42E−1(+)
CIMS-T2	**0**	2.95E+2(+)	**0(≈)**	3.05E+0(+)	1.27E+1(+)	1.09E+1(+)
CILS-T1	2.01E+1	2.12E+1(+)	2.01E+1(≈)	2.01E+1(≈)	2.01E+1(−)	2.01E+1(−)
CILS-T2	3.40E+3	3.91E+3(+)	3.44E+3(≈)	**3.27E+3(≈)**	3.70E+3(≈)	4.09E+3(+)
PIHS-T1	**1.28E+2**	2.81E+2(+)	1.32E+2(≈)	1.55E+2(+)	1.58E+2(+)	1.74E+2(+)
PIHS-T2	**2.39E−11**	2.04E−10(≈)	1.85E−4(+)	5.33E−2(+)	4.66E−2(+)	3.48E−2(+)
PIMS-T1	1.86E−1	**2.93E−2(≈)**	4.94E−1(+)	1.26E+0(+)	1.59E+0(+)	2.06E+0(+)
PIMS-T2	5.18E+1	4.16E+2(+)	3.84E+1(≈)	3.69E+1(≈)	3.57E+1(≈)	**2.95E+1(≈)**
PILS-T1	**2.57E−14**	3.81E−1(+)	2.24E−1(+)	5.25E−11(+)	1.67E−1(+)	4.40E−8(+)
PILS-T2	**1.99E−18**	2.44E+0(+)	5.60E−1(+)	6.85E−10(+)	3.16E−1(+)	1.82E−5(+)
NIHS-T1	4.76E+1	4.88E+3(+)	4.61E+1(≈)	4.57E+1(≈)	3.85E+1(≈)	**3.77E+1(+)**
NIHS-T2	**7.45E−11**	2.86E+2(+)	4.38E+0(≈)	1.35E+1(+)	2.39E+1(+)	1.77E+1(+)
NIMS-T1	**1.90E−3**	2.96E−3(−)	2.62E−3(+)	5.02E−3(+)	6.50E−3(+)	7.37E−3(+)
NIMS-T2	**5.50E+0**	9.96E+0(+)	6.00E+0(≈)	6.35E+0(+)	8.64E+0(+)	8.86E+0(+)
NILS-T1	**1.16E+2**	3.29E+2(+)	1.46E+2(+)	1.78E+2(+)	2.42E+2(+)	2.93E+2(+)
NILS-T2	**3.58E+3**	4.13E+3(+)	3.78E+3(≈)	4.18E+3(+)	4.75E+3(+)	5.01E+3(+)
Number of + / ≈ / −		15 / 2 / 1	8 / 10 / 0	13 / 5 / 0	14 / 3 / 1	16 / 1 / 1
Problem	**SCLMTO**	**SCLMTO-ktp(0.6)**	**SCLMTO-ktp(0.7)**	**SCLMTO-ktp(0.8)**	**SCLMTO-ktp(0.9)**	**SCLMTO-ktp(1.0)**
CIHS-T1	**0**	5.35E−3(+)	4.18E−3(+)	7.65E−3(+)	1.26E−2(+)	4.00E−2(+)
CIHS-T2	**0**	1.91E+1(+)	1.42E+1(+)	1.77E+1(+)	2.62E+1(+)	3.81E+1(+)
CIMS-T1	**1.49E−10**	1.05E+0(+)	1.29E+0(+)	1.17E+0(+)	2.21E+0(+)	2.12E+0(+)
CIMS-T2	**0**	2.33E+1(+)	2.97E+1(+)	2.94E+1(+)	7.66E+1(+)	7.32E+1(+)
CILS-T1	2.01E+1	2.01E+1(−)	2.01E+1(−)	2.00E+1(−)	**2.00E+1(−)**	2.02E+1(+)
CILS-T2	3.40E+3	4.39E+3(+)	4.79E+3(+)	4.98E+3(+)	5.37E+3(+)	6.07E+3(+)
PIHS-T1	**1.28E+2**	2.07E+2(+)	1.72E+2(+)	2.27E+2(+)	2.14E+2(+)	2.95E+2(+)
PIHS-T2	**2.39E−11**	2.01E−2(+)	4.91E−2(+)	1.15E−1(+)	7.17E−1(+)	1.55E+1(+)
PIMS-T1	1.86E−1	1.72E+0(+)	2.29E+0(+)	2.27E+0(+)	2.61E+0(+)	3.08E+0(+)
PIMS-T2	5.18E+1	4.64E+1(≈)	3.88E+1(≈)	4.02E+1(≈)	3.98E+1(≈)	8.46E+2(+)
PILS-T1	**2.57E−14**	7.66E−2(+)	2.97E−1(+)	1.46E−1(+)	7.58E−2(+)	5.36E−1(+)
PILS-T2	**1.99E−18**	8.29E−2(+)	2.79E−1(+)	1.38E−1(+)	1.02E−1(+)	6.47E−1(+)
NIHS-T1	4.76E+1	3.83E+1(+)	5.01E+1(+)	4.17E+1(+)	6.04E+1(+)	9.69E+1(+)
NIHS-T2	**7.45E−11**	1.75E+1(+)	1.24E+1(+)	2.25E+1(+)	4.57E+1(+)	7.08E+1(+)
NIMS-T1	**1.90E−3**	1.35E−2(+)	2.11E−2(+)	2.99E−2(+)	5.98E−2(+)	2.84E−1(+)
NIMS-T2	**5.50E+0**	7.87E+0(+)	9.69E+0(+)	1.18E+1(+)	1.23E+1(+)	1.21E+1(+)
NILS-T1	**1.16E+2**	3.47E+2(+)	4.77E+2(+)	5.32E+2(+)	6.74E+2(+)	7.78E+2(+)
NILS-T2	**3.58E+3**	5.28E+3(+)	6.04E+3(+)	6.60E+3(+)	7.98E+3(+)	8.71E+3(+)
Number of + / ≈ / −		16 / 1 / 1	16 / 1 / 1	16 / 1 / 1	16 / 1 / 1	18 / 0 / 0